What is in `embeddings.npz`?

Aviral Garg · Forgis screening task · 10 Oct 2026

Every number here comes out of `code/analyze.py` (numpy/scipy/torch, no sklearn, seed 0, about 4 minutes on CPU). Raw values are in `out/results.json`.

1. Identification

Given: `T` (2500, 512) f32, `I` (2500, 512) f32, `Y` (2500,) i64. No other metadata.

- `Y` is a class label: 31 distinct values in [1, 34], 31–33 absent ⇒ a subset of a 34-class dataset. 49–150 samples/class; majority class 6.0%.
- Not ℓ_2-normalised: $\|\mathtt{T}\| = 9.06 \pm 0.86$, $\|\mathtt{I}\| = 10.19 \pm 0.46$. Raw encoder output, taken before any normalisation layer.
- Two different encoders: per-dim variance correlation 0.0017, mean per-dim correlation 0.18, and only 4 of 512 dims reach $|r| > 0.5$. They share no basis.
- `T`'s inputs repeat exactly, `I`'s never do: `T` 2075/2500 unique (173 duplicate groups, max multiplicity 30), `I` 2500/2500, and 98.8% of duplicate groups sit inside one class. 83% of `T` rows are still unique, so this is a pool with class-confined repeats.
- Row order carries nothing: $\cos(\mathtt{I}_i, \mathtt{I}_{i+1}) = 0.6579$ against 0.6581 for a random pair, and the same at lags 2, 5, 10 and 50. `Y` has 2416 runs in 2500 rows.

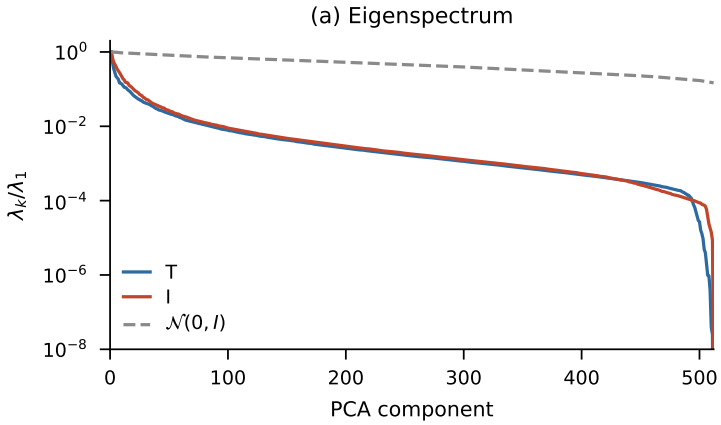
Hypothesis: a CLIP-style dual encoder, with `T`=text, `I`=image and `Y`=class.

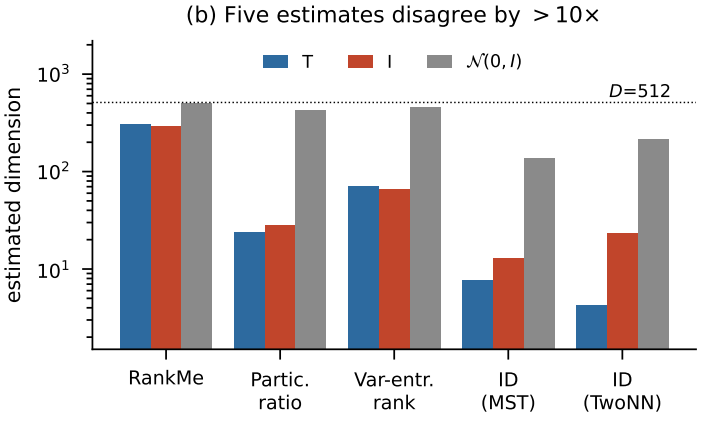
- For: D=512 is CLIP ViT-B/32's shared width for both towers; the vectors are unnormalised, as a pre-ℓ_2 dump would be; two towers explain the disjoint bases; repeated captions explain `T`'s duplicates; the mean-offset (§4) matches CLIP's documented gap.
- Against / open: paired similarity barely exceeds same-class similarity (§4), so the pairing is class-level, and 34 classes weren't enough to pin down the source dataset. A hypothesis with its evidence, nothing stronger.
- Decisive test, which needs the source data: encode known image/caption pairs with ViT-B/32 and compare norms, gap and spectrum. I have not measured reference CLIP norms.

It is not a HEPA dump. HEPA (Petersen et al., arXiv:2605.11130) emits a (h_t, h^*) pair per sample, so I tested for that. Three checks, all of them negative:

- D=512 ≠ HEPA's d=256 (their §3.1).
- No temporal structure, but HEPA's pairs index $(t, \Delta t)$ over episodes.
- HEPA's representations are ℓ_2-normalised onto the unit sphere (appendix A.2, and the Loss row of appendix E: "L1 on L2-normalised"), whereas $\|\mathtt{T}\| = 9.06$, $\|\mathtt{I}\| = 10.19$. I first used a SIGReg-distance test here; it does not discriminate, because HEPA applies SIGReg to the predictor output $\hat{h}$ and never to h_t or h^*, so a genuine dump need not look isotropic.

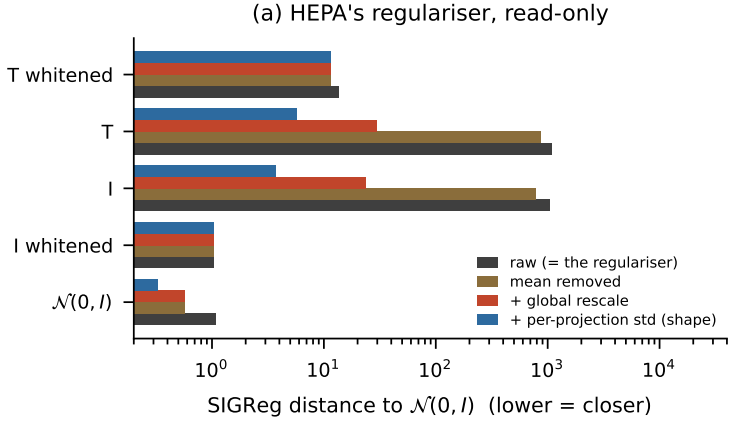
2. Geometry: every rank estimate disagrees

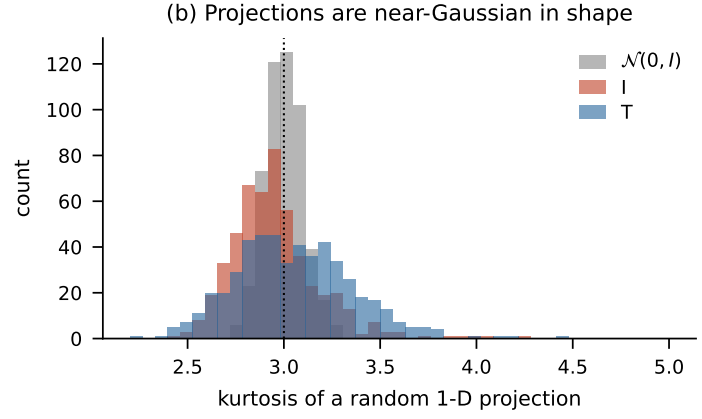




- Strongly anisotropic: 90% of variance needs 110 (`T`) / 93 (`I`) of 512 dims, after which the spectrum falls five orders of magnitude.
- Five standard estimates span > 10× on the same data (`T`/`I`): RankMe 309/292, participation ratio 24/28, variance-entropy rank 71/66, MST-ID 7.7/12.9, TwoNN-ID 4.3/23.2, and they disagree on the ordering: TwoNN puts `T`≪`I`, RankMe puts `T`>`I`.
- On an isotropic Gaussian control both ID estimators badly underestimate (136, 216 vs. a true 512), so at N=2500, D=512 absolute ID values are untrustworthy. I compare only within one estimator and name it every time.

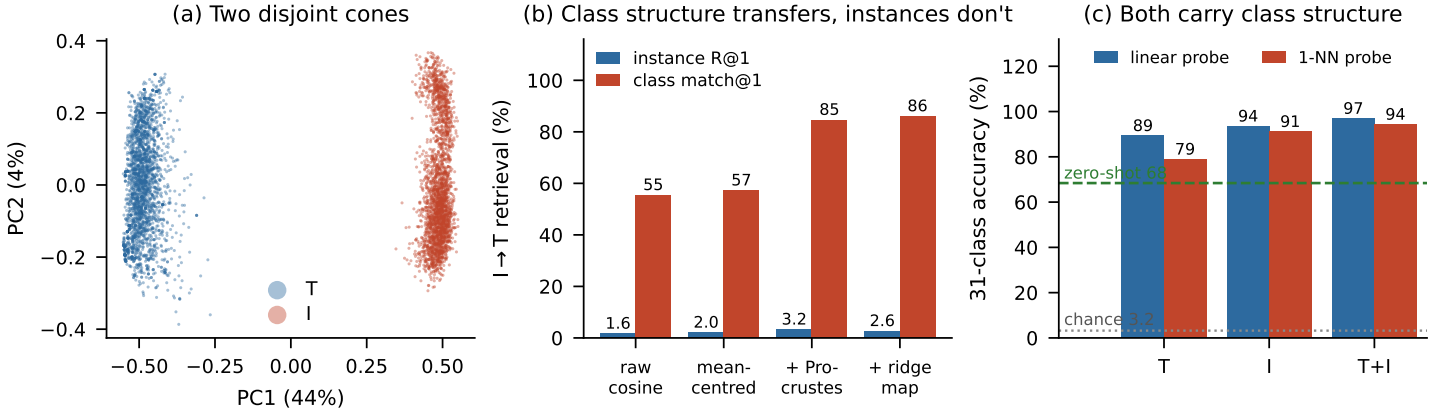
3. Isotropy, measured with HEPA's own regulariser





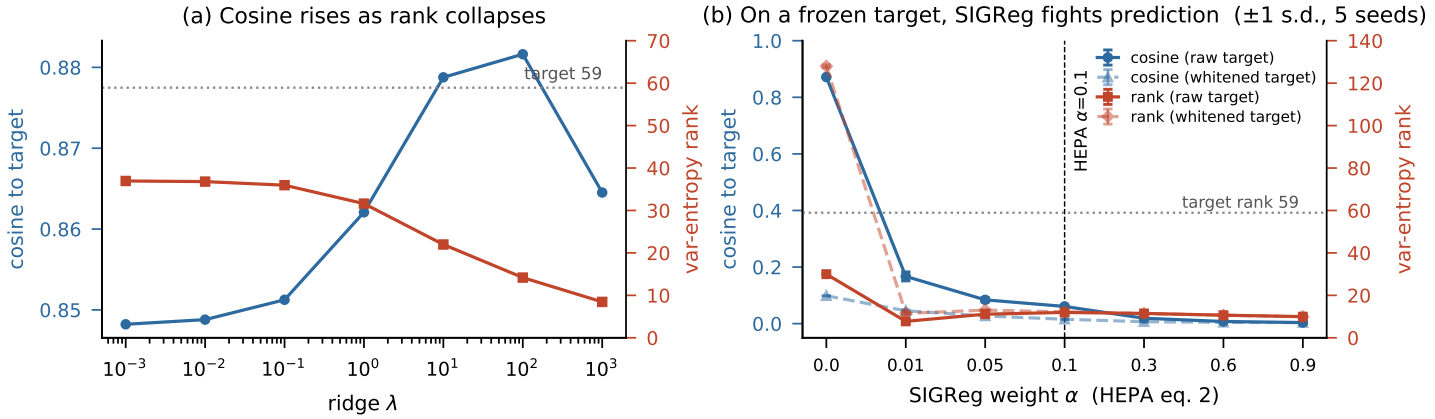
- raw is LeJEPA algorithm 1 verbatim (arXiv:2511.08544): random unit directions, no centring, no rescaling, Epps–Pulley with window $e^{-t^2/2}$, 17-knot trapezoid on $[-5, 5]$. Only that line is the regulariser; the next three strip one factor each (`T` / `I` / $\mathcal{N}(0, I)$):
  - raw 1091 / 1045 / 1.08 · mean removed 860 / 781 / 0.57 · + one global rescale 29.8 / 23.9 / 0.57 · + per-projection standardised 5.7 / 3.7 / 0.33.
- Almost all of the departure is scale and anisotropy; the shape is already near-Gaussian. Mean projection variance 0.044 / 0.070 where SIGReg wants 1.0; kurtosis 3.06 / 2.95 against 3.00. Drives §5.
- Whitening `T` makes it worse: shape-only 5.7→11.4, kurtosis 3.06→4.15, because whitening amplifies near-null directions into heavy tails. Whitening `I` behaves (1.04), so a blanket "whiten first" step isn't safe here.

4. The modality gap is a rotation for class structure, not for instances



- Protocol: centring, rotation and ridge map are fitted on the 2000 training rows; retrieval is queried with the 500 held-out rows against the full 2500-row gallery, chance R@1 0.04%. The cloud statistics and panel (a) are descriptive over all 2500 rows, nothing fitted. An earlier version fitted and scored on all 2500 rows: 1.6%→17.6% at instance level against the 3.2% below, class match 88.2% vs 84.6%. The finding survives, narrowed.
- Disjoint clouds: $\cos(\bar{\mathtt{T}}, \bar{\mathtt{I}}) = 0.319$, $\|\bar{\mathtt{T}} - \bar{\mathtt{I}}\| = 0.97$ (normalised); PC1 alone separates them.
- Raw cosine barely sees the pairing: paired 0.2781, same-class 0.2715, random 0.2215. The 0.007 margin says the correspondence is class-level.
- The rotation transfers at class level and barely at instance level (`I`→`T`, held-out): class match@1 55.4%→84.6%, but instance R@1 only 1.6%→3.2% (80× chance), R@5 7.2%→10.8%. Mean-centring alone gives 2.0% / 57.2%. That split is the finding, with its null: if the map recovered class structure only and within-class ranking were uninformative, R@1 would be ≈ class@1 × 31/2500 = 1.0%. I measure 3.2%, so real instance signal exists, about 3× that null.
- Zero-shot (held-out `I` classified by train-fitted class-mean `T` prototypes): 68.4% → 89.0% after the same rotation; chance 3.2%.
- Not a pure rotation: an unconstrained ridge map edges class match (86.0%), loses instance R@1 (2.6%), and has singular values far from flat (mean 0.22, s.d. 0.34).
- Both carry class structure: linear probe 89.4 / 93.6 / 97.0% and 1-NN 79.0 / 91.2 / 94.4% (`T` / `I` / concat). The linear−kNN gap is +10.4 pts on `T` but +2.4 on `I`, so `T`'s class information is linearly decodable while being less metrically organised. Repeated captions would do exactly that.

5. Prediction in embedding space: cosine improves, nothing downstream does



- Setup: the obvious JEPA experiment here. Context `T`, target `I`, predict in representation space. Held-out metrics; MLP results are mean ± s.d. over 5 seeds.
- A ridge map looks better and better while collapsing. Over $\lambda : 10^{-3} \to 10^{2}$, cosine rises 0.848→0.882 and R^2 0.13→0.34, while the prediction's variance-entropy rank falls 37.0→14.2 against a target rank of 58.9. Had I reported cosine alone it would have hidden a 2.6× rank collapse.
- But rank does not predict the downstream number either. Over that same range held-out retrieval is flat (R@1 8.0, 7.8, 7.8, 8.2, 8.4, 7.8%), falling to 4.2% only at λ=10^3. Scored against the 500-row held-out gallery, so chance is 0.2% and these are not comparable with §4's 2500-row numbers. So cosine improves monotonically while buying nothing downstream, and rank tracks the collapse but not the task. Report all three.
- SIGReg on the predictor does not fix it here. 2-layer MLP under HEPA eq. (2), $\mathcal{L} = (1-\alpha)\|\hat{h} - h^*\|_1 + \alpha\mathcal{L}_{\text{SIGReg}}$: at α=0, cosine 0.8716±0.0004 / rank 30.0±0.1; at HEPA's own α=0.1, cosine 0.0609±0.0044 / rank 11.9±0.2. Every $\alpha \geq 0.01$ destroys prediction, with seed spread far below the effect.
- A scale mismatch is involved, but it is not what separates me from HEPA. SIGReg wants unit per-projection variance; my frozen target sits at 0.070, about 14× off. HEPA's representations are on the unit sphere (appendix E): at d=256 that gives 1/256 = 0.0039, about 256× off, so it is further from SIGReg's target than mine. Normalising is not the difference. Control: against a train-whitened target (projection variance 1.00 on the fitting rows, 1.38 held-out, against 0.07 raw) the mismatch is largely gone and prediction still collapses (cosine 0.109±0.000→0.048±0.003 at α=0.01; rank 129±1→11.4±0.6). What is left is the target being frozen.
- What this does and does not say about HEPA. HEPA's target is never frozen: weight-shared and jointly trained (§3.1), and their appendix I.3 shows the method is robust to an EMA target instead (τ=0.99; joint wins 6/12, loses 4, ties 2, all within ±0.11 h-AUROC), choosing joint training "for its simplicity". Both leave the target free to move toward the Gaussian SIGReg wants, and even τ=0.99 is far more mobile than frozen. I measured the frozen-target regime, which neither setting in I.3 covers: there, no α bought both prediction and rank. That gap is where my proposal starts.

6. Caveats

- One 80/20 split throughout, seed 0. Every fitted object in §4 and §5 (centring, rotation, ridge map, prototypes, probes, the whitening transform, the MLP) is fitted on the 2000 training rows and scored on the 500 held out. I don't resample it, so the split is an untested source of variance. MLP results are 5 seeds; the ridge sweep is exact. §2–§3 describe all 2500 rows and fit nothing.
- The CLIP identification is unconfirmed (test named in §1), and the α sweep is one architecture at one budget (2-layer MLP, 200 epochs) on unnormalised targets: I'd stand behind the direction, not the crossover point, and I have not run HEPA itself.